

Pap-smear cervical cytology image dataset

Overview

This dataset contains 3,000 Pap smear microscopy images collected at The Ocean Road Cancer Institute (ORCI), Tanzania, between April and October 2025.

Data Format

- Modality: Pap smear microscopy images
- Capture device: Digital microscope (10×/0.25 objective; integrated camera)
- File format: JPEG (RGB)

Folder structure

The dataset is organized into 8 class folders (Bethesda System categories):

- ASC-US– Atypical Squamous Cells of Undetermined Significance
- ASC-H – Atypical Squamous Cells, cannot exclude HSIL
- AGC – Atypical Glandular Cells
- HSIL – High-grade Squamous Intraepithelial Lesion
- LSIL – Low-grade Squamous Intraepithelial Lesion
- NILM – Negative for Intraepithelial Lesion or Malignancy
- SC – Squamous Cell Carcinoma
- A – Adenocarcinoma

File naming convention

Images are named as: *class_abbreviation_z*, where *z* is a sequential number within each class.

Example: ASC-US_1

How to access

Download and unzip the folder

Ethics and de-identification

All images are provided without patient identifiers. Use of the dataset should comply with relevant ethical and institutional guidelines.